Royal Berkshire NHS Foundation Trust

Diabetes in pregnancy guideline (GL983)

Approval

Approval Group	Job Title, Chair of Committee	Date
Maternity & Children's Services Clinical Governance Committee	Chair, Maternity Clinical Governance Committee	14th June 2024

Change History

Version	Date	Author, job title	Reason
4.0	May 2024	A Brooks, Consultant Obstetrician L Critcher/J Eustace, Diabetes Specialist MWs K Foteini, Consultant Endocrinologist	Reviewed and updated to reflect current practice and national recommendations.
4.1	Aug 2024	J Eustace, Diabetes Specialist MW A Brooks, Consultant Obstetrician	Live change to amend criteria for OGTT to clarify gestation it should be done.

To comply with NG121 (published Mar 2019) and QS192 (published Sept 2019) staff need to ensure that;

- They clarify with women and birthing people with obstetric complications or who have received no antenatal care whether or how they would like their birth companion(s) involved in discussions about care during labour and birth and review this regularly.
- They involve women and birthing people in planning their care by asking about her preferences and expectations for labour and birth.
- They take into account previous discussions, planning decisions and choices and keep the women or birthing person and their companion(s) fully informed.

To be read in conjunction with

- Neonatal hypoglycaemia GL359

CONTENTS

1.0 Management of pre-existing Diabetes (Type 1 and Type 2)

1.1 Antenatal Management

1.1.1 GP/CMW should refer women to the Diabetes Specialist Midwives (DSM) **as soon as the pregnancy is confirmed**.

1.1.2 DSM remains the first point of contact throughout the pregnancy and after delivery while in the hospital. All pregnant women with DKA, whatever gestation, should be discussed with the Obstetric on call team. Advice from the endocrine team should be sought for women admitted via obstetrics with DKA.

1.1.3 Complications of Diabetes in pregnancy are miscarriage, congenital anomalies, stillbirth, pre-eclampsia, macrosomia (increased risk of shoulder dystocia), increased operative delivery and admission to neonatal unit, birth injury or increased perinatal mortality and morbidity. The children of type 2 diabetic mothers are at increased risk of diabetes and obesity themselves.

1.1.4 All women requiring treatment of retinopathy, those with CKD 2 + PCR >30 or CKD 3 (unless associated with pre-eclampsia) should be discussed via the regional maternal medicine meeting. Patients are referred via OARS to Silver Star in Oxford, please discuss with Bryn Kemp or Alex Brooks.

1.1.5 Adverse outcomes are increased in women with diabetes who develop pre-eclampsia or fetal growth restriction. If pre-eclampsia is diagnosed please organise a growth scan if this has not happened within the previous 2 weeks and >24/40.

1.1.6 Women and birthing people with pre-existing diabetes will be seen in a multidisciplinary clinic with a Consultant Obstetrician, Consultant Endocrinologist and a DSM in attendance.

1.1.7 Access to a specialist diabetic dietician is available via fortnightly Microsoft Teams drop-in sessions.

Scans in the pregnancy

1.1.8 Viability scan at 7-8 weeks, routine dating and anomaly scans

1.1.9 Foetal echocardiogram due to diabetes at 22-24 weeks is no longer indicated as cardiac views form part of the routine anomaly scan.

1.1.10 Foetal growth monitored from 28 weeks every four weeks until 36 weeks. Weekly or two weekly scanning may be required in some cases

Medications

1.1.11 Women should be on 5mg of Folic acid pre-pregnancy and continue to take it till 14 weeks of the pregnancy

1.1.12 Women should be commenced on 150mg of Aspirin from 12 weeks onwards (unless advised to do so earlier) to reduce the risk of early onset pre-eclampsia and to stop at 36 weeks.

1.1.13 Antihypertensive like ACE inhibitors and ARB (Angiotensin receptor blockers), cholesterol lowering agents (statins), and all diabetes medications except insulin and metformin should be discontinued if planning pregnancy or as quickly as possible after a positive pregnancy test.

Glucose control

1.1.14 Ensure women with diabetes (type 2) are given a blood glucose meter. Women will be asked to monitor their blood glucose levels six times a day. These should be before breakfast, lunch, dinner, and one hour after breakfast, lunch and dinner. The glycaemic target should be

- Fasting (pre breakfast) less than 5.3mmol/l
- Pre lunch and dinner – 4.0 – 6.0mmol/l
- Post-meal readings should be aimed at <7.8mmol/l one hour after food and <6.4mmol/l 2 hours after food
- Pre bed (at least 2 hours after dinner) less than 6.4mmol/l

1.1.15 All members of the MDT have access to the GDM app where women with type 2 diabetes and GDM record blood glucose results.

1.1.16 In instances where women are, not achieving good blood glucose control or they have poor hypo awareness the MDT will write to the woman's GP to prescribe Libre.

1.1.17 For women with type 1 diabetes - discuss and offer real time continuous glucose monitoring (CGM) to replace intermittent glucose monitoring. Flash monitoring may be available at the discretion of primary care to women with type 2 diabetes taking insulin. Any women that chose CGM and who are not previously familiar with it, should have education and training provided on the product by the DSM.

1.1.18 Hypoglycaemic safety to be discussed with the woman and she should be given Hypo-kits and an adult relative should be taught how to recognise symptoms of hypoglycaemia and treat at home in emergencies.

1.1.19 Ensure that all women with type 1 diabetes have a ketone monitor and in date test strips available to them.

1.1.20 If women are concerned with reduced foetal movements after 26/40 they should be advised to ring triage on **0118 322 7304**. Women known to have pre-eclampsia or fetal growth restriction are known to be at significantly increased risk of adverse outcomes if there is co-existing diabetes.

Blood Tests in Pregnancy

1.1.21 In addition to the booking bloods, a baseline urea, Creatinine, electrolytes, HbA1c and thyroid function tests to be done at the initial visit and every trimester. PCR (protein: creatinine ratio) to be sent if persistent proteinuria of >/= 2 on two separate occasions.

1.1.22 Consider referral to the renal team or Silver Star if booking creatinine >120 micromol/L OR PCR >30 OR if eGFR <45ml/min/1.73 m^2.

1.1.23 **HbA1c Targets (Saving Babies Lives v.3)**

1.1.24 Measure HbA1c levels in all pregnant women with pre-existing diabetes at the booking appointment to determine the level of risk for the pregnancy.

1.1.25 Fructosamine may be considered as an alternative in women where the laboratory advises HbA1c measurement is not possible (some Haemoglobin variants, e.g., sickle cell anaemia)

1.1.26 Women with an HbA1c >86 mmol/mol are advised to avoid pregnancy and counselling should acknowledge this.

1.1.27 HbA1c measurement should be repeated in all women with pre-existing diabetes at 24+0 -30+0 weeks, aiming for an HbA1c <48 mmol/mol

Green	HbA1c 43 mmol/mol or less	Continue current care
Amber	HbA1c 44-48 mmol/mol	Consider additional input to improve glucose management
Red	HbA1c more than 48 mmol/mol	MDT discussion required. Offer additional input to improve glucose management including alternative methods of monitoring treatment. Offer increased fetal surveillance, and re-discuss increased risk of stillbirth, birth and neonatal complications

Retinal Screening

1.1.28 Women will be advised to have retinal screening once in every trimester, this will be organised by the DSM in the first instance. Those requiring treatment during pregnancy should be discussed with the regional maternal medicine centre in Oxford and their control and fetal condition reviewed. Please inform Alex Brooks or Bryn Kemp so an OARS referral can be undertaken to Silver Star (SBL v.3).

1.1.29 Women and birthing individuals with renal impairment will be referred to Oxford University Hospital via OARS.

Thromboprophylaxis

1.1.30 VTE assessment should be done in the antenatal period and repeated during admissions and in the postnatal period

1.1.31 LMWH should be strongly considered in all women with PCR >300 mg/mol

Steroid and Magnesium Sulphate Administration

1.1.32 Magnesium sulphate should be given as per routine protocol for neuroprotection and management of pre-eclampsia / eclampsia

1.1.33 Antenatal steroids for fetal lung maturation should be given as per guideline. Women with diabetes are increased risk of hyperglycaemia following administration of betamethasone. If at home women should contact triage for admission if their blood glucose is >11 mmol/L and they are unable to correct this.

Plan for delivery

1.1.34 Delivery of women with pre-existing diabetes should be organised between 37+0 and 38+6 weeks. If there are medical or Obstetric complications like pre-eclampsia, macrosomia or growth restrictions delivery may have to be considered earlier. The potential for baby to be admitted in Special Care Unit needs to be discussed.

1.1.35 Aim blood sugar <7 mmol/L in the 24hrs before delivery, using a sliding scale if necessary. This reduces the risk of neonatal hypoglycaemia.

1.1.36 A customised delivery plan is made in joint diabetes clinic for all planned deliveries and will be available on EPR.

1.2 Intrapartum Management

1.2.1 All inpatients using their own insulin must be assessed by the admitting midwife for their suitability to self-administer insulin as per the RBH trust guidelines.

1.2.2 If the woman is already using CGM/flash monitoring (e.g., FreeStyle Libre) then check a reported result against a finger prick result. If the results are within 0.3 mmol/L then the existing monitor can be used.

1.2.3 Women will continue with their normal medication (Metformin &/or insulin) until they are in active labour or after ARM. Metformin and short acting insulin should be discontinued in all women who are not eating.

1.2.4 An IV cannula should be sited as soon as the woman is in established labour. A second IV cannula is required if the woman needs Variable Rate Insulin Infusion (VRII).

1.2.5 For women with T1DM offer VRII once in established labour or after ARM. Stop the short acting (bolus) insulin but continue with the long acting (basal) insulin while on VRII. Women may choose to continue with an insulin pump as an alternative.

For women on Metformin /diet controlled Type 2 blood glucose should be monitored every two hours. Increase monitoring to hourly if blood glucose >7.0mmol/l. If BM persistently (>2 readings) > 7.0mmol/l start VRII.

1.2.6 Target blood glucose levels of 4.0mmol – 7.0mmol/l should be maintained during labour.

1.2.7 Monitor blood glucose every hour in the first stage of labour and every half hour for the second stage of labour for insulin dependent diabetics both Types 1 and 2 For women with T1DM using CGM, this device can be used to monitor blood glucose levels – **see flowchart in appendix, noting exclusion criteria.** This includes removal of CGM sensor and insulin pump prior to any procedure in theatre.

1.2.8 Check U&Es 4-6 hourly during labour to maintain potassium and bicarbonate

1.2.9 For VRII see appendix 2. These are usually discontinued after delivery when the patient is eating and drinking normally. If being used antenatally then discontinue once the woman is eating and drinking normally and blood sugars are <8 mmol/L. Please ensure short acting insulin is administered 30 minutes before stopping VRII.

1.3 Postnatal care

1.3.2 Newly delivered T1 and T2 DM women will need significantly smaller doses of insulin in the postnatal period. This should be in the woman's delivery plan on EPR.

1.3.3 If there is no documented plan reduce to the first insulin dose recorded in the pregnancy minus a further 25% OR reduce the last pregnancy dose by 50%. If still unsure contact endocrine registrar on bleep 192/199

1.3.4 Monitor blood glucose four times a day before each meal and before bedtime. Aim for pre-meal glucose levels of 6.0mmol/l to 10.0mmol/l in the immediate postnatal period. For women with T1DM, CGM recordings can be used. If sensor and insulin pump has been removed, this can be replaced immediately in the postnatal period from the women's own kit.

1.3.5 Type 2 DM on Metformin/ diet control may or may not need medication. Most will have their insulin discontinued. Please review the delivery on EPR for postnatal instructions

1.3.6 If BMs persistently >10.0mmol/l consult DSM or endocrine registrar.

1.3.7 If breast feeding/expressing, diet should be adjusted to include 10-15 gr of carbohydrate and to drink fluids with each feed and when expressing

2.0 Management of Gestational Diabetes Mellitus (GDM)

2.1 Screening

Risk factors for Gestational Diabetes Mellitus (GDM) are as follows:

- Previous GDM.
- Ethnicity with a high prevalence of diabetes – South Asian, Black African and Afro-Caribbean. This includes mixed ethnicity.
- Previous baby with birth weight >4500gm
- BMI > 30kg/m2
- First degree relative with diabetes
- Glycosuria of 1+ on 2 occasions or >=2+ on 1 occasion
- Women on anti-psychotics
- Scan showing AC >95th centile with or without polyhydramnios
- Previous bariatric surgery

2.1.1 Referral to be made for all women with risk factors. The diabetes team will organise appropriate testing according to gestation.

2.2 Diagnosis

OGTT is the preferred and validated method of diagnosis.

We will offer this test if the woman is <32/40. From 32wks a fasting blood glucose & H6A.c will be taken

GDM will be diagnosed if the fasting blood glucose is >/= 5.6mmol/l or more and the 2-hour level is >/= 7.8mmol/l.

Women not suitable for OGTT (previous gastric bypass) are offered 1 week blood sugar monitoring at approximately 26/40 via the diabetes midwives and an individual decision made by the diabetes team.

HbA1c >41 mmol/L and <28/40 or >39 mmol/L and >28/40 is used to diagnose gestational diabetes.

HbA1c >48 mmol/L is likely to represent type 2 diabetes that will not resolve after pregnancy.

2.3 Antenatal care

2.3.1 Aim to maintain the fasting blood sugar below 5.3mmol/l and pre-prandial blood glucose levels between 4.0mmol/l and 6.0mmol/l

2.3.2 Offer immediate treatment with insulin, with or without metformin as well as changes in diet and exercise, to women with gestational diabetes who have a fasting plasma glucose level of >/=7.0 mmol/litre at diagnosis.

2.3.3 Consider immediate treatment with insulin, with or without metformin, as well as changes in diet and exercise, for women with gestational diabetes who have a fasting plasma glucose level of between 6.0 and 6.9 mmol/litre if there are complications such as macrosomia or hydramnios.

2.3.4 Aim to keep the one-hour post prandial blood glucose levels below 7.8mmol/l and two-hour post prandial below 6.4mmol/l.

2.3.5 The women will be asked to send readings on a regular basis to the DSMs by email or phone and, where diet/lifestyle changes fail to achieve the above glycaemic levels, will initiate the appropriate oral hypoglycaemic/insulin treatment.

2.3.6 Offer ultrasound scans for growth from 28/40

2.3.7 Deliver between 40 and 40+6 weeks; earlier if complications

2.3.8 For steroid prophylaxis, see Appendix 1

Monitoring HbA1c

- Measure HbA1c levels in all women with gestational diabetes at the time of diagnosis to identify those who may have pre-existing type 2 diabetes.

2.4 Intrapartum care

2.4.1 Patients using their own Insulin must be assessed by the admitting midwife for their suitability to self-administer Insulin prior to labour, as per RBH trust guidelines. Give the woman the leaflet and complete the self-assessment form. Women will continue with their normal diet and normal medication (Metformin and/or Insulin) until such time as they are assessed as being in active labour. Aim to keep blood glucose levels between 4.0mmol/l - 7.0mmol/l during labour.

2.4.2 For all women with GDM take blood glucose minimum 2 hourly and increase to hourly if it is above 7.0mmol/l. If persistently above 7.0mmol/l, then consider using a Variable Rate Insulin Infusion to control blood glucose levels. This is important if delivery is not imminent, in order to prevent neonatal hypoglycaemia. (See Appendix 3).

2.4.3 Women will continue with their normal medication (Metformin &/or insulin) until they are in active labour or after ARM. Metformin and short acting insulin should be discontinued in all women who are not eating

2.5 Postnatal care

The vast majority of women diagnosed with GDM will see their blood glucose levels return to normal within 24-36 hours.

2.5.1 Ask the woman to monitor her blood glucose levels minimum four times daily (pre-meals and pre-bed).

2.5.2 The ward staff may discharge the woman as long as her blood glucose is below 6.0mmol/l, or between 6.0 – 10.0mmol/l with a message left for the DSM to follow up. There is a discharge folder on the ward detailing the process of discharge.

2.5.3 Women should be advised to have a Fasting blood glucose test 5 weeks after the delivery and this should be reviewed by the GP at the 6-week visit. Some GP use an HbA1c at 3 months as an alternative. Referral to the Diabetes Prevention pathway locally should be offered to the women, to be arranged via their GP.

2.5.4 If levels remain above 10.0mmol/l after 24 hours, then the patient must not be discharged without being seen by the DSMs or a member of the Diabetes MDT, as they are likely to have pre-existing diabetes.

(Appendix 1) Steroid Prophylaxis in pregnancy for diabetic women

1. Steroid prophylaxis is indicated as per usual guidelines.

2. Babies of diabetic mothers are at increased risk of respiratory problems after delivery, which may require admission to Buscot. However, although steroids reduce this risk there are increased rates of neonatal hypoglycaemia and a possible association with developmental delay.

3. Steroids usually elicit a rise in maternal blood glucose levels which may be apparent immediately or take some time

4. Women on insulin will be advised to increase their insulin by 10- 20% from steroid administration and for the next 48 hours. The increased dose will be documented in the maternity record on EPR. Please note that women who do not take insulin outside pregnancy are often far less confident about correcting blood sugars.

5. Women who have been given steroids to take at home are advised to check their blood glucose levels measured every four hourly.

   If the BM reading is >/= 11mmol/L repeat test in 1 hour.

   If still >/=11mmol/L call triage and come to the hospital for VRII. If <11.0mmol/L to continue to monitor four hourly.

6. Diabetic ketoacidosis (DKA) is a serious risk for diabetic women receiving steroids and admission may be necessary for those who are Type1/ Metformin/ diet controlled diabetics. (See pages 13-15 for management of DKA)

7. Urine ketone testing should be carried out twice daily if admitted.

   If ketones are:

   1+ in the urine dipstick check capillary ketones

   2+ or more in urine dipstick alert Obstetric registrar.

8. Women who have had steroids and are on VRII **should eat and drink normally** and continue their normal diabetes medication (both short and long acting insulin) in addition to their VRII; if levels are still rising treat by moving to higher VRII algorithm (see appendix 2)

(Appendix 2) Variable rate insulin infusion (VRII) – *Formerly known as Sliding Scale of Insulin*

Indications for use

This may occur during intercurrent illness

Steroid treatment

Intrapartum

Diabetic ketoacidosis in pregnancy **CAUTION - FOR DIABETIC KETOACIDOSIS PLEASE REFER TO THE MANAGEMENT OF DKA SECTION**

General Principles

For women on VRII after steroid administration**.** If there is no nausea/vomiting and the woman is eating and drinking normally please continue the short acting (bolus) insulin and the long acting (basal) insulin that the woman is on, along with the VRII

If the VRII was started to manage blood sugar during the intrapartum period continue on the long acting insulin only along with the VRII. If not eating normally short acting insulin should be avoided.

Algorithm >	**1**	**2**	**3**
For most women	**For women not controlled on algorithm 1 or needing > 80 units/day of insulin**		**For women not controlled on algorithm 2 (after specialist advice)**
CBG Levels (mmol/L)	**Infusion Rate (units/hr = ml/hr)**		
< 4	**STOP INSULIN FOR 20 MINUTES Treat hypo as per guideline (re-check CBG in 10 minutes)**		
4.0 – 5.5	**0.2**	**0.5**	**1.0**
5.6 – 7.0	**0.5**	**1.0**	**2.0**
7.1 – 8.5	**1.0**	**1.5**	**3.0**
8.6 – 11.0	**1.5**	**2.0**	**4.0**
11.1 – 14.0	**2.0**	**2.5**	**5.0**
14.1 – 17.0	**2.5**	**3.0**	**6.0**
17.1 – 20.0	**3.0**	**4.0**	**7.0**
> 20.1	**4.0**		**6.0**

ALL women with diabetes should have Capillary Blood Glucose (CBG) testing hourly whilst on VRIII for the management of steroid hyperglycaemia during pregnancy

Algorithm 1 Most women will start here

Algorithm 2 Use this algorithm for women who are likely to require more insulin (on steroids; on >80 units of insulin during pregnancy; or those not achieving target on algorithm 1)

Algorithm 3 Use this for women who are not achieving target on algorithm 2 (No patient starts here without diabetes or medical review)

If the woman is not achieving targets with these algorithms, contact the diabetes team (out of hours: Medical SpR on call)

Target CBG level = 4 – 7.8 mmol/L

Check CBG every hour whilst on VRIII

Move to the higher algorithm if the CBG is > target and is not dropping

Move to the lower algorithm if CBG falls below 4 mmol/L or is dropping too fast

PREPARATION

The VRII infusion is prepared with soluble insulin (Novorapid) 50 units in 50ml of Normal Saline (Sodium chloride 0.9%) in 50ml IV syringe administered IV through non-returnable valve via a Trust approved infusion pump.

Alongside VRII there should be infusion of 0.45% saline with 5% glucose and 0.15% potassium chloride **OR** an alternative substrate solution to above is 5% glucose (be aware of hyponatremia with this regime) Selection of subsequent fluids should be based on electrolytes and fluid status. *PLEASE PRESCRIBE THIS ON EPR AND DO NOT USE THE POWER PLAN FLUDS*

The fluid should run at 50ml/hr

Continue with the VRII for 24 hours after the second dose of steroid

(Appendix 3) Caesarean sections in diabetic women

ALL INSULIN PUMPS AND BLOOD SUGAR MONITORS (e.g., FreeStyle Libre) MUST BE REMOVED IN THEATRE. Women are advised to bring new sets to use after delivery. These are not available from the diabetes team.

ELECTIVE

- All inpatients using their own insulin must be assessed by the admitting MW for their suitability to self-administer Insulin as per RBH trust guidelines.
- Women may be admitted on the morning of the surgery to the Day assessment Unit (DAU) AT 07:30hrs. In some cases, admission to Iffley ward the day before may be suggested
- Metformin can be taken normally whilst the patient is eating. Short acting insulin should be taken with meals on day before surgery, but long acting insulin may be reduced the night before surgery (for the dose see the diabetes booklet in maternity notes). No Metformin or short acting insulin should be taken on the morning of the elective surgery. Some women may be advised to continue long acting insulin on the morning of surgery. If there is no diabetes plan on EPR (the majority of women with gestational diabetes) then do not give any insulin on the day of surgery.
- Check blood glucose one hour after admission. If the BM >8.0mmol/l check BM 30 mins later. Check blood ketones and if >0.6 mmol/l transfer to delivery suite for management with VRII.
- If the blood glucose is <3.5 mmol/l prior to midnight and patient able to swallow safely, correct hypoglycaemia by fast acting carbohydrates (GlucoGel, Glucotab, Jelly babies, sweet drink which the woman may have) or use the glucogel tab in the HYPO box. Follow up with two slice of toast OR two digestive biscuits.
- If woman unable to co-operate and /or swallow safely use Glucagon injection (in the fridge in delivery suite/ Iffley ward) or 75mls of 20% IV glucose. Inform medical staff and transfer the woman to the DS.
- If VRII is needed for control of blood sugar during the surgery, the anaesthetist will manage it.

EMERGENCY

- To avoid neonatal hypoglycaemia in theatre, aim for blood glucose to be maintained between 4-7 mmol/l.
- If GA is required, then measure BM every half hourly and if regional analgesia is used measure BM every hour.

- Blood glucose will be maintained by the VRII if needed which will be managed by the anaesthetist.

(Appendix 4) Management of the Neonate of the diabetic woman

1. Diabetes in pregnancy carries the risk of hypoglycaemia, respiratory distress syndrome, polycythaemia, jaundice, hypothermia.
2. Babies of mothers who have been on insulin should be admitted to the Transitional Care bay on Marsh (Level 4)
3. Babies of diabetic mothers should be fed immediately after birth. Please consider whether colostrum is available. Encourage frequent feeding and ensure no longer than 3 hours between feeds.
4. Observation of the clinical and neurological status should be documented.
5. If the mother intends to bottle feed start at 10-15mls/kg per feed 3hourly for the first 24 hours. Encourage breastfeeding over formula.
6. Neonatal blood glucose should be checked prior to the second feed and within 4 hours of the birth, unless symptomatic of hypoglycaemia.
7. Blood glucose should be checked as per the Neonatal Hypoglycaemia Prevention & Management (GL359) guideline.
8.
    (i) Routine monitoring of term babies can be discontinued after 2 consecutive normal values. Recommence if concerns of hypoglycaemia.

    (ii) Pre-term babies (<37/40) should be continued for 48hrs.

GL359 Flowchart A1 - Management of Infants at risk of hypoglycaemia (for use with BLOOD GAS blood sugar monitoring)

At birth

- Dry and place the baby skin-to-skin care in a warm draught free room
- Put hat on the baby, and cover with a warm blanket
- Encourage and support early breast feeding within the first hour after birth
- For women who choose to formula feed give 10-15mls/kg within the first hour after birth.
- Provide written and verbal information to parents (Patient Information Leaflet for Hypoglycaemia), see appendix 3

Within 4hours

- Check pre-feed blood glucose level prior to second feed (within 4hrs of birth)

Is blood glucose level more than or equal to 2.0mmol/L?

No → See Flowchart B

Yes

- Encourage frequent feeding and ensure no longer than 3 hours between feeds
- Assess the need for helping the mother with feeding, hand expression, recognition of early feeding cues and signs of effective attachment and feeding
- For women who chose to formula feed, give 10-15mls/kg per feed 3 hourly over the first 24 hours after birth

Within 8hours

- Check pre-feed blood glucose level prior to third feed (within 8hrs of birth)

Is blood glucose level more than or equal to 2.0mmol/L?

No → See Flowchart B

Yes

- Continue to support responsive breast feeding
- If formula fed give 10-15mls/kg per feed 3 hourly over the first 24 hours
- Ensure mother understands how to assess effective feeding and knows how to escalate concerns
- No further blood glucose monitoring required unless there are clinical signs of hypoglycaemia (Box 2)
- Observe feeding in hospital for 24 hours
- Complete at least one recorded breast feeding assessment using local/BFI tool prior to transfer home

See Flowchart B
Blood sugar <1.0 – Bleep neonatal Registrar/ANNP immediately

Box 1. Indications for blood glucose monitoring

- Intrauterine growth restriction (≤2nd centile)
- Infants of Diabetic mothers
- Large birth weight >4.5kg
- Maternal beta-blocker use
- Preterm <37 weeks
- Arterial Cord pH <7.1 BE≥-12
- Hypothermia (Temp <36.5 *)
- Neonatal Sepsis-raised CRP
- Structural problems-Cleft lip or palate
- Known Metabolic Disorders like MCADD

*If baby has a normal blood sugar with a temperature under 36.5C and the temperature responds to intervention the baby does not need to be on the protocol

Weight(kg) ≤2nd centile		
Gestation (weeks)	Boys	Girls
36	1.9	1.8
37	2.1	2.0
38	2.3	2.2
39	2.5	2.45
40	2.65	2.6
41	2.8	2.75
42	2.9	2.85

Box 2. Signs that may indicate hypoglycaemia

- Lethargy
- Abnormal feeding behaviour especially after a period of feeding well
- High pitched cry
- Altered consciousness
- Hypothermia (<36.5°c)
- Hypotonia
- Seizures
- Grunting
- Cyanosis
- Apnoea

GL359 Flowchart B1 - Management of Infant with blood glucose 1.0-2.0 on BLOOD GAS and no abnormal clinical signs

Does the baby have clinical signs consistent with hypoglycaemia (see box 1)?

Yes →

See Flowchart C
Bleep neonatal Registrar/ANNP immediately

No ↓

- Support breast feeding (see Box 2)
- If mother is choosing to formula feed, aim to
- give 10-15ml/kg feed and continue 3 hourly
- Consider administration of 40% buccal dextrose gel 200mg/kg (Appendix 1)
- Recheck blood glucose before next feed (3 hours)

Is the blood glucose level >2.0 mmol/l?

No →

Any of:
- Blood glucose <1.0mmol/L
- 2 consecutive blood sugars <2.0mmol/L
- 3 blood glucose levels

Yes ↑ (See Flowchart C)

No ↖ (Does the baby have clinical signs consistent with hypoglycaemia?)

Yes ↓

- Give feed: breast feed and/or offer expressed breast milk (3-5ml/kg)
- In formula fed infants aim for 80-100ml/kg/day in 3 hourly feed volumes
- Recheck blood glucose before next feed (3 hours)

Is the blood glucose level >2.0mmol/l?

No ↗

Yes ↓

- Continue to support breast feeding and feed responsively
- After 2 consecutive pre-feed BG measurements >2.0mmol/l (or **48 hours if born <37 weeks**) discontinue BG monitoring unless there are abnormal clinical signs (Box 1), in which case use Flowchart C
- Observe feeding for at least 24 hours
- Complete at least one breastfeeding assessment tool prior to transfer home

Box 1. Signs that may indicate hypoglycaemia
- Lethargy
- Abnormal feeding behaviour especially after a period of feeding well
- High pitched cry
- Altered consciousness
- Hypothermia (<36.5^{o}c)
- Hypotonia
- Seizures
- Grunting
- Cyanosis
- Apnoea

Box 2. Supporting breast feeding
- Encourage continuous skin-to skin contact
- Teach mother to hand express breast milk
- Give colostrum obtained to baby by the method suitable to parents
- Continue to encourage hand expression at least 8-10 times in 24 hours and support feeding on the breast until infant is feeding effectively

GL359 Flowchart C - Management of symptomatic hypoglycaemia, prolonged hypoglycaemia, and/or blood glucose <1.0 on BLOOD GAS

Symptomatic (see box 1) hypoglycaemia is an emergency: bleep neonatal registrar immediately for urgent paediatric review
Admit baby to neonatal unit for urgent intervention

Registrar **Bleep 138**
SHO **Bleep 333**

For recurrent or persistent hypoglycaemia, contact paediatric team for review, likely to need neonatal admission
May need investigations
Consider increased feed frequency or volume, high energy feed or NG feeds but default treatment as below or if concerns

Box 1. Signs that may indicate hypoglycaemia

- Lethargy
- Abnormal feeding behaviour especially after a period of feeding well
- High pitched cry
- Altered consciousness
- Hypothermia (<36.5°c)
- Hypotonia
- Seizures
- Grunting
- Cyanosis
- Apnoea

- Obtain IV access
- Consider screening and treating sepsis
- Obtain lab sample to confirm blood glucose, consider full hypoglycaemia screen (see appendix 4), and check ketones (blood ketone meter or put on urine bag)

Unable to obtain immediate IV

Dextrose gel can be given while IV access is obtained OR intramuscular glucagon 200micrograms/kg

Treat IV with 10% glucose 2.5ml/kg
Start IV infusion of 10% glucose at 60-90ml/kg/d

- Do no stop the establishment of breast feeding unless the baby is too sick to feed or there is a clinical contraindication to feeding. Support expression of breast milk
- In formula fed babies continue feeds if no contraindication to enteral feeding
- Recheck blood glucose on blood gas after 30 minutes

Blood glucose <1.0 on blood gas or symptomatic hypoglycaemia:
Confirm hypoglycaemia screen completed and/or repeat
Give IV 10% glucose 2.5ml/kg
Increase glucose delivery rate by increasing volume and/or concentration of glucose infusion (see appendix 5)

Blood glucose 1.0-2.5 on blood gas with no abnormal clinical signs:
Increase glucose delivery rate by increasing volume and/or concentration of glucose infusion (see appendix 5)
Continue to feed if no contraindication

Blood glucose >2.5 on blood gas:
Slow wean of IV fluids
Continue enteral feeding
Continue to monitor blood glucose until infant is on full enteral feeds and blood glucose values are >2.0mmol/l for at least 24 hours

Repeat blood glucose every 30 minutes until blood glucose >2.0 on blood gas

3.0 Management of complications of Diabetes in pregnancy

3.1 Diabetic Ketoacidosis (DKA)

3.1.1 DKA needs to be treated as a medical emergency. All patients' with T1D should have blood ketone meters and testing strips at home. Please invite any patient with a blood ketone >1.5 mmol/L to attend immediately whatever their gestation.

3.1.2 DKA in adults with T1D is associated with a mortality rate of 5-10%. In pregnancy, the stillbirth rate is 160 per 1000births. Ketones are toxic to the foetus

3.1.3 It can occur in T2D and in gestational diabetes

3.1.3 Possible causes are Infection, hyperemesis gravidarum, neglect of diabetes care, steroid prophylaxis, faulty glucose meter or insulin pumps, poor compliance

3.1.4 DKA in pregnancy may present with normal/ modest rise in glucose levels.

3.1.5 Symptoms include nausea, vomiting, abdominal pain, polyuria, polydipsia and leg cramps, dehydration, blurred eyesight, tachypnoea, leg cramps, distinct smell in the breath and coma

3.1.6 Any woman known to have diabetes who contacts triage/MAU feeling unwell should be invited in for assessment at whatever gestation

3.2 Diagnosis of DKA

3.2.1 Presence of diabetes mellitus (any type of diabetes) DKA in pregnancy can occur in a woman with known diabetes with normal blood glucose AND

- Urinary ketones 2+ or Blood ketones >3.0mmol/l (high risk 1.5mmol/l) AND
- Blood gas pH <7.3 and/or serum bicarbonate <15mmol/L. Use venous gas
- Ask when they last ate and when they had their last insulin
- If on insulin pump, discontinue pump and start FRII

3.3 Investigations

3.3.1 Bloods should be sent for FBC, U&Es, glucose and ketones, bicarbonate, venous lactate, cultures if indicated, G&S ABG to be considered if the SPo2 is low

3.3.2 Urine for MC&S

3.3.3 Consider Chest X-ray and ECG

3.3.3 CTG as soon as woman is stabilised

3.4 Treatment

3.4.1 Assessment ABC

3.4.2 Site 2 IV cannulae both grey (16G)

3.4.3 Inform obstetric registrar, obstetric consultant, and anaesthetic consultant early

3.4.4 Request Medical review and endocrine registrar (bleep number 199/192)

3.4.5 Assess for severity and consider management in the ITU/HDU

3.4.6 15-minute recording of BP and Pulse, hourly recording of respiration and MOWS and 4 hourly temperatures.

3.4.7 Commence IV fluids **as soon as possible** through the first cannula as follows:

Fluid	Volume	Duration	Potassium
0.9% Sodium chloride	1 litre	1 hour	None
Give 500ml bolus if SBP<90mmHg (can be repeated)			
0.9% Sodium chloride	1 litre	2 hour	Replace as below
0.9% Sodium chloride	1 litre	2 hours	Replace as below
0.9% Sodium chloride	1 litre	4 hours	Replace as below
0.9% Sodium chloride	1 litre	6-8 hours	Replace as below

Potassium replacement chart (Aim potassium level of 4.0-5.5mmol/l)

Potassium level	KCl per litre of fluid	Comment
<3.5mmol/l	40mmol	Refer to endocrine registrar/outreach team
3.5-5.5mmol/l	20mmol	
>5.5mmol/l	No Potassium	
Potassium chloride should Not be infused at a rate faster than 10mmol/hr		

Commence FRIII (Fixed rate Intravenous Insulin infusion) as below

Set up an insulin infusion of 50 units of soluble insulin (Humulin S) or Actrapid insulin in 49.5ml of 0/9% NaCl via syringe driver

Weight	Hourly Infusion rate
60-69kg	6 units
70-79kg	7 units
80-89kg	8 units
90-99kg	9 units
100-109kg	10 units
110-119kg	11 units
120-129kg	12 units
130-139kg	13 units
>/=140kg	14 units
If the patient is on Long acting insulin (e.g., Glargine (Lantus, Toujeo, Abasaglar), Determir (Levemir), Degludec (Tresiba) continue on usual dose and time If patient is not meeting the treatment target with above rule out insulin infusion malfunction Insulin infusion rate can be adjusted depending on the response and can be increased by 1ml/hr if the target is not met and there are no malfunctions in the pump	

To avoid hypoglycaemia, add 10% Dextrose to run along normal saline when the blood glucose is </=14mmol/L @ 125ml/hr (or 250ml/hr if blood glucose <7ml/hr)

Treatment Targets

- Drop in ketones at 0.5ml/hr at least
- Rise in bicarbonate by at least 3ml/hr
- Fall in glucose by at least 3ml/hr
- Restoration of the circulating blood volume
- Maintain potassium in normal range

Monitoring

- First 6 hours: blood glucose and ketones at least hourly
- Blood gas at 1,2,4,6,8 hours to check pH and potassium
- Monitor fluid status as needed
- The foetus should be monitored continuously and abnormalities in the FHR may improve with improvement of maternal condition
- Assess clinically for signs of fluid overload and cerebral oedema

Signs of resolution of DKA

- Blood ketones of <0.3mmol/L and pH >7.3
- If ketones are <0.3mmol/L ketone monitoring can be discontinued unless blood glucose >11.0mmol/L
- Aim to convert to subcutaneous insulin when DKA is resolved, and the patient is able to eat and drink
- If DKA is resolved but the patient cannot eat, and drink switch fixed rate insulin infusion to variable rate infusion and continue with the basal insulin and start 5% Dextrose to avoid hypoglycaemia if blood glucose <14.0mmol/l (Please see Appendix 3)

Hypoglycaemia Avoidance

- Hypoglycaemia occurs when the blood glucose is <4.0mmol/l
- It may present with headache, sweating, blurred vision, dizziness/shaking, palpitation, anxiety, hunger, weakness and fatigue, feeling faint, slurred speech and confusion.

(Appendix 5) Management of Hypoglycaemia

Management of Hypoglycaemia